Home Work 1 Solution

1. a). $(1001.1101)_2$

$(1001)_2 = 2^3.1 + 0.2^2 + 0.2^1 + 1.2^0$

$= 9$

$(.1101)_2 = 2^{-1} + 2^{-2} + 2^{-4}$

$= .8125$

$= (9.8125)_{10}$

b). $(8.45)_{10}$

$8/2 = 4 + 0$

$4/2 = 2 + 0$

$2/2 = 1 + 0$

$1/2 = 0 + 1/2$

$(8)_{10} = (1000)_2$

$.45(2) = 0 + .9$

$.9(2) = 1 + .8$

$.8(2) = 1 + .6$

$.6(2) = 1 + .2$

$.2(2) = .4$

$(.45)_{10} = .0111\overline{00}$

$(8.45)_{10} = 1000.01\overline{1100}$

2. There will be two case.

I. $a_{n+1} = 0$ ((HOP)

$x - f(l)x = (0.0 \ldots a_{n+1} \ldots).2^e \leq (0.00 \ldots 0111 \ldots).2^e$

$= (1 \times 2^{-(n+2)} + 2^{-(n+3)} + \ldots).2^e$

$= 2^{-(n+2)}(1 + \frac{1}{2} + \ldots).2^e$

$= 2^{-n-1+e}$

Now,

$$-\epsilon = \frac{x - f(x)}{x} \leq \frac{2^{-n+1+e}}{x}$$

$$\leq \frac{2^{-n+1+e}}{(0.1)_2 2^e}$$

$$= \frac{2^{-n-1+e}}{2^{e-1}}$$

$$= 2^{-n} \implies -\epsilon \leq 2^{-n}$$

$$\epsilon \geq 2^{-n}$$

Case II. ROUND UP

$$x - f(x) \geq -(0.00\ldots1)\times 2^e = -2^{-(n+1)}\cdot 2^e$$
$$= -2^{-n-1+e}$$

$$-\epsilon = \frac{x - f(x)}{x} \geq \frac{-2^{-n-1+e}}{x}$$
$$\geq \frac{-2^{-n-1+e}}{(0.1)2^e}$$
$$= \frac{-2^{-n-1+e}}{2^{e-1}}$$
$$= -2^{-n}$$

$$\therefore\ -\epsilon \geq -2^{-n} \implies \epsilon \leq 2^{-n}$$

Hence $-2^{-n} \leq \epsilon \leq 2^{-n}$

Proved

3. $$\text{Rel}\, f(x) = \frac{f(x_T) - f(x_A)}{f(x_T)}$$

$$f(x_T) - f(x_A) = f'(\xi)(x_T - x_A)$$

where ξ is between x_T and x_A $\implies \xi \approx x_T \implies x_A \approx x_T$

So, $$f(x_T) - f(x_A) \approx \frac{1}{2\sqrt{x_T}}(x_T - x_A)$$

$$\implies \text{Rel}\, f(x) \approx \frac{x_T - x_A}{2\sqrt{x_T}\cdot\sqrt{x_T}}$$
$$= .5\,\frac{(x_T - x_A)}{x_T}$$
$$= .5\,\text{Rel}(x)$$

$$\implies \boxed{\text{Rel}\, f(x) = \frac{1}{2}\text{Rel}(x)}$$

4. Given: $f(n) = \sin(arn) - \sin(a)$

Then, $f(n) = 2\cos\left(\frac{arn+a}{2}\right)\sin\left(\frac{arn-a}{2}\right)$

$= 2\cos\left(\frac{2arn}{2}\right)\sin\left(\frac{n}{2}\right)$

$f(n) = 2\cos\left(\frac{2arn}{2}\right)\sin\left(\frac{n}{2}\right)$